AUTOMATED DATA HEALTH & PROFILING DASHBOARD

Group 18

Revanth S (CB.AI.U4AID25043) • G Kamalesh (CB.AI.U4AID25014)
Bhanu Prakash K (CB.AI.U4AID25025) • Elango Megabala G (CB.AI.U4AID25013)

Abstract

In modern data-driven ecosystems, the integrity and performance of Artificial Intelligence models depend heavily on the quality of introductory data representations and execution pipelines. This paper presents the architecture of a pure software solution: the Automated Data Health & Profiling Dashboard, designed to streamline data pre-processing and exploratory profiling. Developed using an interactive web framework built entirely in Python, the system enables users to dynamically ingest semi-structured and raw CSV or JSON datasets. The backend engine automatically intercepts structural anomalies and deploys robust cleaning modules capable of handling missing values through statistical imputation strategies or conditional row exclusion, while simultaneously automating mathematical text transformations such as categorical one-hot encoding. Crucially, the platform maps out data behavior by calculating core descriptive metrics from scratch, including central tendency, variance, and feature relationships through a live covariance matrix. Distribution anomalies are flagged via kurtosis computations, paired with an automated five-point summary visualized dynamically through box-and-whisker plots. By abstracting the mathematical complexity of data validation into an intuitive browser interface, this open-source tool functions as a fully deterministic software agent that optimizes the foundational data science process pipeline. Evaluated across diverse data configurations, the system reduces structural preparation overhead, ensuring high-fidelity mathematical datasets are delivered rapidly to downstream diagnostic models.

References

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